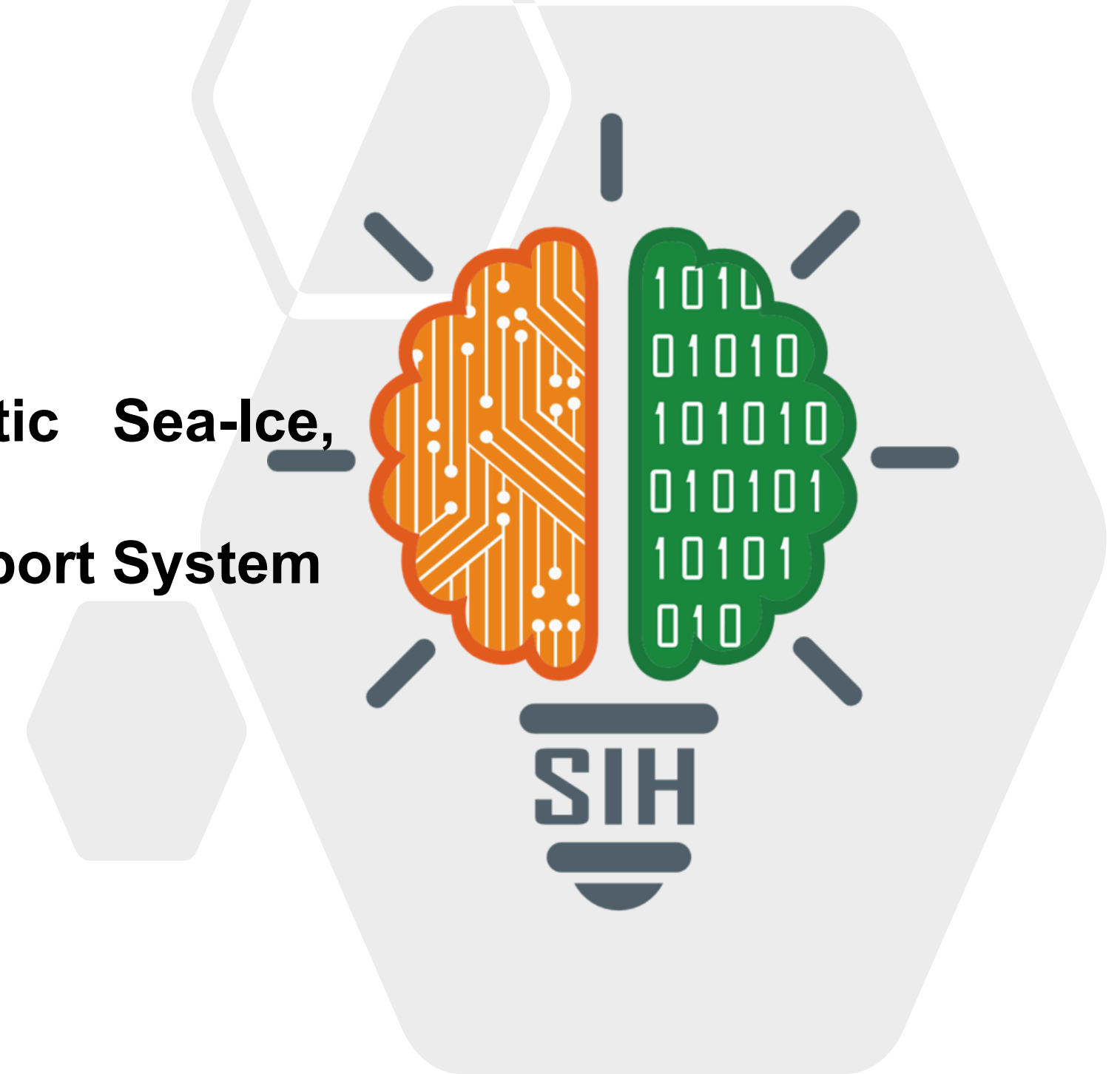
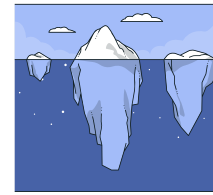


ICE GAURD

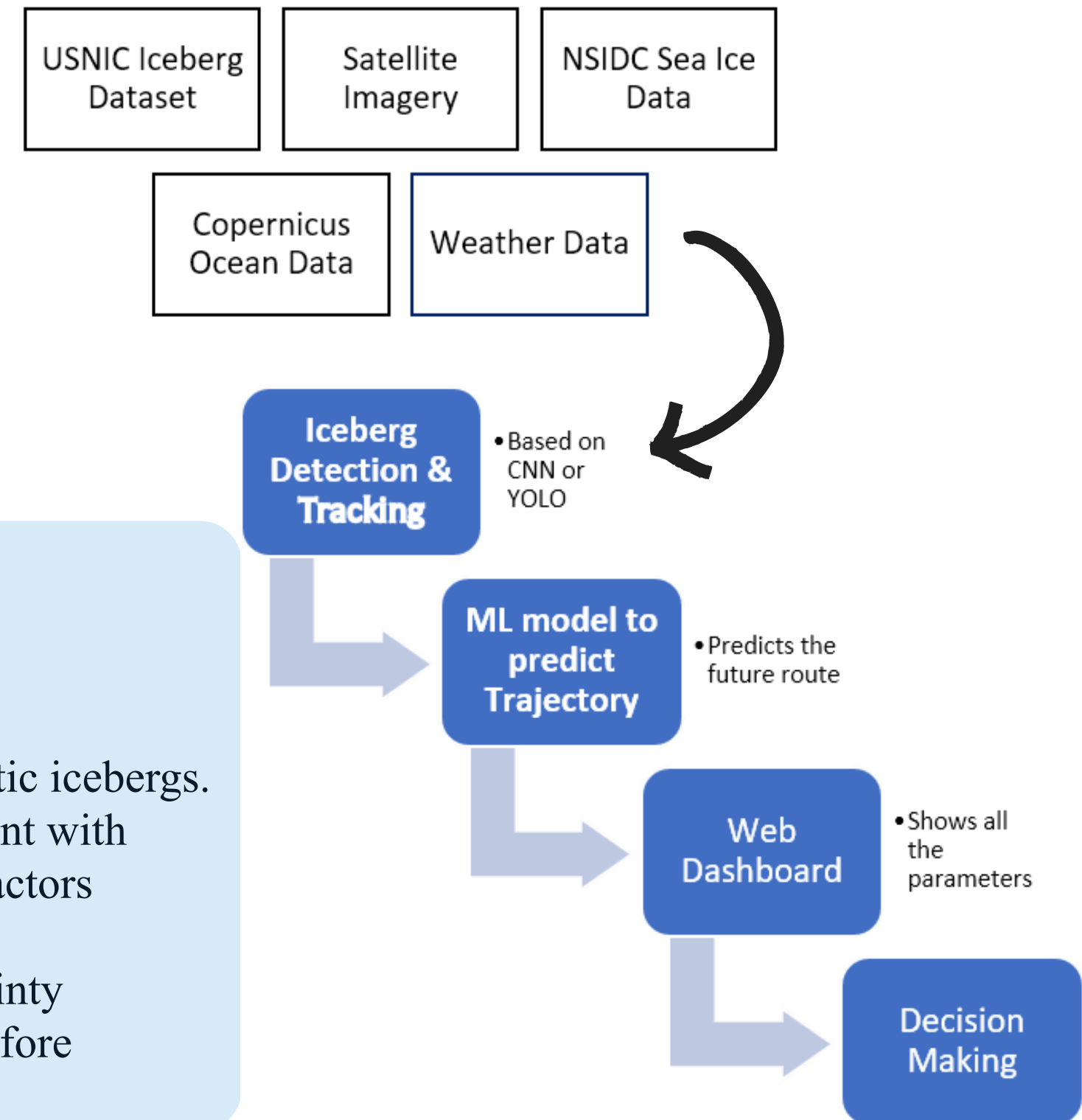
- Problem Statement ID – 26059
- Problem Statement Title- AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System
- Theme- Transportation & Logistics
- PS Category- Software
- Team ID-
- Team Name - Zero Hour





PROPOSED SOLUTION :

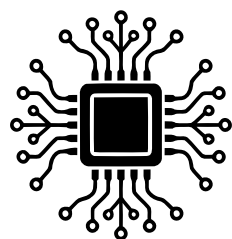
- Collect historical iceberg positions from **USNIC**.
- Obtain Antarctic sea-ice concentration from **NSIDC**.
- Use **computer vision** to detect icebergs from satellite imagery.
- Train ML models such as **XGBoost/LSTM** to predict future iceberg positions.
- There will be an interactive dash board measuring the environmental parameters, sea-ice conditions.



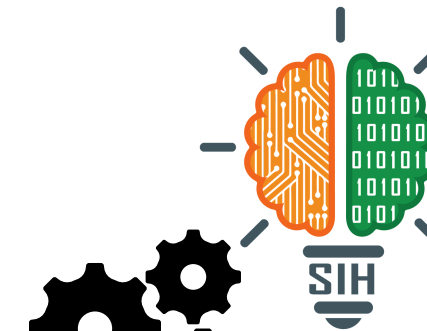
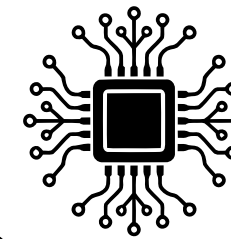
HOW IT ADDRESSES THE PROBLEM:

- **Real-time iceberg monitoring:** Integrates satellite imagery and existing iceberg observations to continuously identify and track Antarctic icebergs.
- **Accurate trajectory prediction:** Combines historical iceberg movement with ocean currents, wind, sea-ice concentration, and other environmental factors to predict future iceberg positions.
- **Early risk identification:** Generates predicted trajectories and uncertainty zones so potentially dangerous iceberg movements can be identified before they intersect with vessel routes

ZERO HOUR



TECHNICAL APPROACH



**SMART INDIA
HACKATHON
2026**

KEY TECHNICAL HIGHLIGHTS:

- We had fused different types of Datasets to make an unified geospatial dataset.
- Uses CNN or YOLO to detect the Iceberg and locate.
- On the basis of historical data ,it will determine its trajectory.
- After that It will be observed in the Web Analytics Part and routes of the Vessels will be optimized from South Gangotri or Moitri Antarctic Point.

Tech Stack



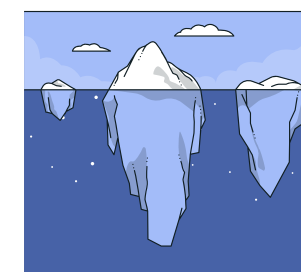
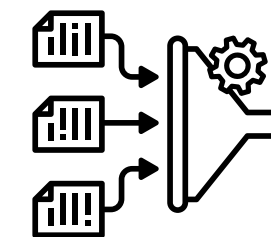
USNIC Iceberg
Dataset

Satellite
Imagery

NSIDC Sea Ice
Data

Copernicus
Ocean Data

Weather Data



Web DashBoard

Trajectory Planning

Data Processing

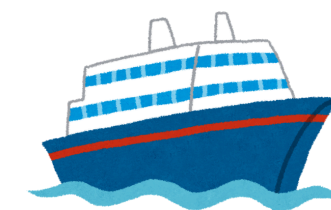
ML Model To Predict

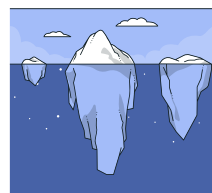
Previous iceberg positions

- > Velocity & direction
- > Ocean currents
- > Wind
- > Sea-ice concentration
- > Other environmental variables



Risk Assessment





ANALYSIS OF THE FEASIBILITY OF THE IDEA

- **Technically feasible:** The project can be implemented using existing satellite imagery, USNIC iceberg data, NSIDC sea-ice data, environmental datasets, and established AI/ML techniques such as CNN, XGBoost, and LSTM.
- **Data feasibility:** Historical iceberg positions and sea-ice observations provide sufficient information for developing and testing prediction models, while satellite imagery enables continuous/near-real-time iceberg monitoring.
- **AI feasibility:** Computer vision can be used for iceberg detection, while ML models can learn relationships between iceberg movement and environmental factors to predict future trajectories.
- **System feasibility:** The proposed interactive dashboard can integrate detected icebergs, predicted trajectories, sea-ice conditions, environmental parameters, and risk zones into a single decision-support platform.

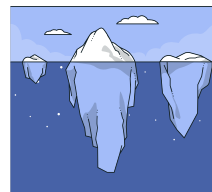
Overall: The idea is highly feasible as a prototype decision-support system, although operational deployment for real-world navigation would require extensive validation and higher-quality real-time data.

POTENTIAL CHALLENGES AND RISKS

- **Data quality:** Different sources may have inconsistent or missing data.
- **Detection accuracy:** Clouds, sea ice, and low-resolution imagery can affect iceberg detection.
- **Trajectory uncertainty:** Iceberg movement depends on changing ocean currents, wind, and sea-ice conditions.
- **Limited training data:** Insufficient labelled Antarctic imagery may affect model accuracy.
- **Prediction risk:** Incorrect predictions could result in inaccurate navigation-risk assessments.

STRATEGIES FOR OVERCOMING THESE CHALLENGES

- **Data preprocessing:** Clean, standardize, and validate data from multiple sources.
- **Improve detection:** Use transfer learning and data augmentation with diverse satellite imagery.
- **Multi-source prediction:** Combine iceberg history with wind, ocean currents, and sea-ice data.
- **Uncertainty modelling:** Generate confidence scores and uncertainty zones for predicted trajectories.
- **Continuous validation:** Test predictions against historical data and continuously improve the models.



IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE

- **Safer navigation:** Helps vessel operators identify potential iceberg hazards in advance.
- **Better decision-making:** Provides predicted iceberg trajectories, sea-ice conditions, and risk information in one platform.
- **Improved situational awareness:** Enables continuous monitoring of changing Antarctic conditions.
- **Reduced uncertainty:** Provides early warnings and uncertainty zones for potential iceberg–vessel encounters.

POTENTIAL IMPACT ON THE TARGET AUDIENCE

1. SAFER NAVIGATION



Helps vessel operators identify potential iceberg hazards in advance, reducing the risk of collisions and ensuring safer voyages.

2. BETTER DECISION-MAKING



Provides predicted trajectories, sea-ice conditions, and risk information in one platform, enabling informed and timely navigation decisions.

3. IMPROVED SITUATIONAL AWARENESS



Enables continuous monitoring of changing Antarctic conditions, helping users stay aware of evolving iceberg and environmental situations.

4. REDUCED UNCERTAINTY



Provides early warnings and uncertainty zones for potential iceberg–vessel encounters, reducing uncertainty in risky environments.

5. OPERATIONAL EFFICIENCY



Helps in route optimization and risk assessment, leading to efficient operations, less delays, and better resource utilization.

6. SUPPORTS CRITICAL OPERATIONS



Supports research missions, supply operations, and tourism activities in Antarctica by enhancing safety and providing reliable decision support.

BENEFITS OF THE SOLUTION

- **Social:** Improves maritime safety and reduces the risk to crew and vessels operating in Antarctic waters.
- **Economic:** Helps reduce potential vessel damage, delays, fuel consumption, and operational costs through safer route planning.
- **Environmental:** Supports safer and more efficient navigation, potentially reducing unnecessary fuel consumption and associated emissions.
- **Scientific:** Enables better understanding of iceberg movement and Antarctic sea-ice dynamics through integrated data and AI analysis.
- **Operational:** Provides a centralized decision-support dashboard for monitoring icebergs, sea ice, environmental conditions, and navigation risk.

ECONOMIC BENEFITS



Reduces operational costs, fuel consumption, delays, and potential damages.

ENVIRONMENTAL BENEFITS



Supports efficient navigation, reduces fuel usage and greenhouse gas emissions.

SOCIAL BENEFITS



Enhances maritime safety, protects crew and vessels, and supports safer operations.

SCIENTIFIC BENEFITS



Improves understanding of iceberg movement, sea-ice dynamics, and Antarctic environmental changes.

OPERATIONAL BENEFITS



- Centralized monitoring and real-time updates
- Early risk identification and alerts
- Better decision-making and route planning

Provides a centralized platform for monitoring icebergs, sea ice, environmental conditions, and navigation risk.

Dataset links :

USNIC Antarctic Iceberg Data:

<https://data.marine.copernicus.eu/products>

Antarctic Sea-Ice Concentration:

<https://usicecenter.gov/Products/>

NSIDC-0051 — Historical Sea-Ice Concentration

<https://nsidc.org/data/g10033/versions/1>

<https://nsidc.org/data/nsidc-0051/versions/2>

Climatic Data :

<https://cds.climate.copernicus.eu/datasets>

Sentinel-1 SAR — For iceberg detection

<https://dataspace.copernicus.eu/data-collections/copernicus-sentinel-missions/sentinel-2>

- **ML Forecasting:** <https://www.sciencedirect.com/science/article/pii/S0048969724049532>
- **Data Fusion:** <https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2024JH000183>

